

## 1. Executive Summary

|                         |        | Traffic Summary        |  | Count      |  |
|-------------------------|--------|------------------------|--|------------|--|
| Metric                  | Value  | Total Flows Observed   |  | 28767      |  |
| Total Threats Mitigated | 48     | Total Packets Analyzed |  | 3110250994 |  |
| High Priority           | 19     | Malicious Packets      |  | 3097770773 |  |
| Low Priority            | 29     | True Negatives Passed  |  | 25127      |  |
| Manual Releases         | 0      |                        |  |            |  |
| Manual Blocks           | 0      | Attack Vector          |  | Count      |  |
|                         |        | ICMP Flood             |  | 6          |  |
| ML Metrics              | Value  | SYN Flood              |  | 18         |  |
| False Positives         | 24     | UDP Flood              |  | 10         |  |
| FP Rate                 | 0.10%  | Uncertain              |  | 14         |  |
| IF Precision            | 99.17% |                        |  |            |  |
| IF Recall               | 93.48% | Action                 |  | Count      |  |
| IF F1-Score             | 96.24% | Quarantined            |  | 10         |  |
| IF Accuracy             | 99.20% | Time Ban               |  | 10         |  |
|                         |        | Blackhole              |  | 0          |  |
|                         |        | Released               |  | 7          |  |

## 2. Performance Benchmark

### 2a. Isolation Forest: Anomaly Detection

| Metric                    | Value  | Description                                          |
|---------------------------|--------|------------------------------------------------------|
| Precision                 | 99.17% | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 93.48% | Share of actual attacks successfully flagged         |
| F1-Score                  | 96.24% | Balanced measure combining Precision and Recall      |
| Accuracy                  | 99.20% | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.10%  | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 6.52%  | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 93.48% | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 99.90% | Normal traffic correctly identified as safe          |

|                | Predicted: Attack | Predicted: Normal |
|----------------|-------------------|-------------------|
| Actual: Attack | TP 2866           | FN 200            |
| Actual: Normal | FP 24             | TN 24927          |

### 2b. Random Forest: Attack Classification

| Metric                    | Value   | Description                                          |
|---------------------------|---------|------------------------------------------------------|
| Precision                 | 100.00% | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 100.00% | Share of actual attacks successfully flagged         |
| F1-Score                  | 100.00% | Balanced measure combining Precision and Recall      |
| Accuracy                  | 100.00% | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.00%   | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 0.00%   | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 100.00% | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 100.00% | Normal traffic correctly identified as safe          |

|           | Predicted: SYN | Predicted: ICMP | Predicted: UDP |
|-----------|----------------|-----------------|----------------|
| Act: SYN  | 1158           | 0               | 0              |
| Act: ICMP | 0              | 964             | 0              |
| Act: UDP  | 0              | 0               | 744            |

### 2c. Hybrid Cascaded

| Metric                    | Value  | Description                                          |
|---------------------------|--------|------------------------------------------------------|
| Precision                 | 99.17% | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 93.48% | Share of actual attacks successfully flagged         |
| F1-Score                  | 96.24% | Balanced measure combining Precision and Recall      |
| Accuracy                  | 99.20% | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.10%  | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 6.52%  | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 93.48% | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 99.90% | Normal traffic correctly identified as safe          |

|                | Predicted: Attack | Predicted: Normal |
|----------------|-------------------|-------------------|
| Actual: Attack | TP 2866           | FN 200            |
| Actual: Normal | FP 24             | TN 24927          |

2d. Response Latency

| Metric                   | Value    | Description                                                             |
|--------------------------|----------|-------------------------------------------------------------------------|
| Detection Time           | 534.6 ms | Average time taken to detect an attack                                  |
| Mitigation Response Time | 16.2 ms  | Interval between the anomaly flag and the FlowMod blocking rule install |

2e. Controller Resource Overhead

| Metric                         | Value | Description                                                           |
|--------------------------------|-------|-----------------------------------------------------------------------|
| Controller CPU (Baseline)      | 3.85% | Ryu controller CPU usage under normal traffic                         |
| Controller CPU (Active Attack) | 8.12% | Ryu controller average CPU usage during a DDoS simulation             |
| Controller CPU (Mitigation)    | 5.50% | Ryu controller CPU usage during simultaneous detection and mitigation |

3. Offences Summary

| Source IP | Sessions | Max Offences | First Seen          | Last Seen           |
|-----------|----------|--------------|---------------------|---------------------|
| 10.0.0.17 | 3        | 3            | 2026-09-25 13:03:50 | 2026-09-25 13:09:46 |
| 10.0.0.18 | 3        | 3            | 2026-09-25 13:03:50 | 2026-09-25 13:07:01 |
| 10.0.0.19 | 3        | 3            | 2026-09-25 13:03:47 | 2026-09-25 13:06:58 |
| 10.0.0.21 | 3        | 3            | 2026-09-25 13:03:47 | 2026-09-25 13:09:52 |
| 10.0.0.22 | 2        | 3            | 2026-09-25 13:03:47 | 2026-09-25 13:06:48 |
| 10.0.0.16 | 3        | 2            | 2026-09-25 13:07:13 | 2026-09-25 13:09:43 |
| 10.0.0.24 | 2        | 1            | 2026-09-25 13:08:05 | 2026-09-25 13:09:48 |
| 10.0.0.20 | 1        | 1            | 2026-09-25 13:09:34 | 2026-09-25 13:09:44 |

| Source IP | Sessions | Max Offences | First Seen          | Last Seen           |
|-----------|----------|--------------|---------------------|---------------------|
| 10.0.0.23 | 1        | 1            | 2026-09-25 13:09:34 | 2026-09-25 13:09:45 |
| 10.0.0.10 | 1        | 1            | 2026-09-25 13:07:22 | 2026-09-25 13:07:32 |
| 10.0.0.2  | 1        | 1            | 2026-09-25 13:03:48 | 2026-09-25 13:03:58 |
| 10.0.0.25 | 1        | 1            | 2026-09-25 13:09:40 | 2026-09-25 13:09:51 |
| 10.0.0.3  | 1        | 1            | 2026-09-25 13:07:20 | 2026-09-25 13:07:31 |
| 10.0.0.4  | 1        | 1            | 2026-09-25 13:07:20 | 2026-09-25 13:07:31 |

## 4. Chronological Mitigation Log

| Timestamp           | Source IP | Class | Vector     | Confidence | Priority | Action        |
|---------------------|-----------|-------|------------|------------|----------|---------------|
| 2026-09-25 13:03:47 | 10.0.0.19 | DDoS  | SYN Flood  | 93.1%      | Critical | Quarantined   |
| 2026-09-25 13:03:47 | 10.0.0.22 | DDoS  | UDP Flood  | 82.5%      | High     | Time Ban (1m) |
| 2026-09-25 13:03:47 | 10.0.0.22 | DDoS  | UDP Flood  | 82.5%      | High     | Time Ban      |
| 2026-09-25 13:03:47 | 10.0.0.21 | DDoS  | UDP Flood  | 83.4%      | High     | Time Ban (1m) |
| 2026-09-25 13:03:47 | 10.0.0.21 | DDoS  | UDP Flood  | 83.4%      | High     | Time Ban      |
| 2026-09-25 13:03:50 | 10.0.0.18 | DDoS  | SYN Flood  | 93.1%      | Critical | Quarantined   |
| 2026-09-25 13:03:50 | 10.0.0.17 | DDoS  | SYN Flood  | 93.3%      | High     | Time Ban (1m) |
| 2026-09-25 13:03:50 | 10.0.0.17 | DDoS  | SYN Flood  | 93.3%      | High     | Time Ban      |
| 2026-09-25 13:03:58 | 10.0.0.19 | DDoS  | SYN Flood  | 93.1%      | Critical | Time Ban (1m) |
| 2026-09-25 13:03:58 | 10.0.0.2  | DDoS  | Uncertain  | 58.9%      | Low      | Released      |
| 2026-09-25 13:03:58 | 10.0.0.19 | DDoS  | SYN Flood  | 93.1%      | Critical | Time Ban      |
| 2026-09-25 13:04:01 | 10.0.0.18 | DDoS  | SYN Flood  | 93.1%      | Critical | Time Ban (1m) |
| 2026-09-25 13:04:01 | 10.0.0.18 | DDoS  | SYN Flood  | 93.1%      | Critical | Time Ban      |
| 2026-09-25 13:04:48 | 10.0.0.22 | DDoS  | UDP Flood  | 82.5%      | High     | Time Ban (2m) |
| 2026-09-25 13:04:48 | 10.0.0.21 | DDoS  | UDP Flood  | 83.4%      | High     | Time Ban (2m) |
| 2026-09-25 13:04:51 | 10.0.0.17 | DDoS  | SYN Flood  | 93.3%      | High     | Time Ban (2m) |
| 2026-09-25 13:04:58 | 10.0.0.19 | DDoS  | SYN Flood  | 93.1%      | Critical | Time Ban (2m) |
| 2026-09-25 13:05:01 | 10.0.0.18 | DDoS  | SYN Flood  | 93.1%      | Critical | Time Ban (2m) |
| 2026-09-25 13:06:48 | 10.0.0.22 | DDoS  | UDP Flood  | 82.5%      | High     | Time Ban (5m) |
| 2026-09-25 13:06:48 | 10.0.0.21 | DDoS  | UDP Flood  | 83.4%      | High     | Time Ban (5m) |
| 2026-09-25 13:06:51 | 10.0.0.17 | DDoS  | SYN Flood  | 93.3%      | High     | Time Ban (5m) |
| 2026-09-25 13:06:58 | 10.0.0.19 | DDoS  | SYN Flood  | 93.1%      | Critical | Time Ban (5m) |
| 2026-09-25 13:07:01 | 10.0.0.18 | DDoS  | SYN Flood  | 93.1%      | Critical | Time Ban (5m) |
| 2026-09-25 13:07:13 | 10.0.0.16 | DDoS  | SYN Flood  | 90.4%      | Critical | Quarantined   |
| 2026-09-25 13:07:24 | 10.0.0.16 | DDoS  | SYN Flood  | 90.4%      | Critical | Time Ban (1m) |
| 2026-09-25 13:07:24 | 10.0.0.16 | DDoS  | SYN Flood  | 93.3%      | Critical | Time Ban      |
| 2026-09-25 13:07:31 | 10.0.0.4  | DDoS  | Uncertain  | 59.1%      | Low      | Released      |
| 2026-09-25 13:07:31 | 10.0.0.3  | DDoS  | Uncertain  | 59.1%      | Low      | Released      |
| 2026-09-25 13:07:32 | 10.0.0.10 | DDoS  | Uncertain  | 68.7%      | Low      | Released      |
| 2026-09-25 13:07:42 | 10.0.0.20 | DDoS  | UDP Flood  | 85.5%      | High     | Time Ban (1m) |
| 2026-09-25 13:07:42 | 10.0.0.20 | DDoS  | UDP Flood  | 85.5%      | High     | Time Ban      |
| 2026-09-25 13:08:05 | 10.0.0.24 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-25 13:08:12 | 10.0.0.23 | DDoS  | ICMP Flood | 96.3%      | High     | Time Ban (1m) |
| 2026-09-25 13:08:12 | 10.0.0.23 | DDoS  | ICMP Flood | 96.3%      | High     | Time Ban      |
| 2026-09-25 13:08:12 | 10.0.0.25 | DDoS  | ICMP Flood | 96.9%      | High     | Time Ban (1m) |
| 2026-09-25 13:08:12 | 10.0.0.25 | DDoS  | ICMP Flood | 96.9%      | High     | Time Ban      |
| 2026-09-25 13:08:15 | 10.0.0.24 | DDoS  | ICMP Flood | 91.1%      | High     | Time Ban (1m) |

| Timestamp           | Source IP | Class | Vector     | Confidence | Priority | Action        |
|---------------------|-----------|-------|------------|------------|----------|---------------|
| 2026-09-25 13:08:15 | 10.0.0.24 | DDoS  | ICMP Flood | 91.1%      | Critical | Time Ban      |
| 2026-09-25 13:08:24 | 10.0.0.16 | DDoS  | SYN Flood  | 93.3%      | Critical | Time Ban (2m) |
| 2026-09-25 13:09:34 | 10.0.0.20 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-25 13:09:34 | 10.0.0.23 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-25 13:09:35 | 10.0.0.17 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-25 13:09:40 | 10.0.0.25 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-25 13:09:41 | 10.0.0.21 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-25 13:09:48 | 10.0.0.24 | DDoS  | Uncertain  | -          | Low      | Released      |
| 2026-09-25 13:09:51 | 10.0.0.25 | DDoS  | Uncertain  | -          | Low      | Released      |
| 2026-09-25 13:09:52 | 10.0.0.21 | DDoS  | Uncertain  | -          | Low      | Released      |
| 2026-09-25 13:10:02 | 10.0.0.22 | DDoS  | Uncertain  | -          | Low      | Quarantined   |

### 5. IP Attack History (Completed Sessions)

| Source IP | Vector     | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                    |
|-----------|------------|----------|-------|----------|---------|----------|----------|---------------------|-------------------------------------------|
| 10.0.0.21 | Uncertain  | 0.0000   | -     | Low      | Phase 1 | 1        | 11s      | 2026-09-25 13:09:52 | Attack Stopped                            |
| 10.0.0.25 | Uncertain  | 0.0000   | -     | Low      | Phase 1 | 1        | 11s      | 2026-09-25 13:09:51 | Attack Stopped                            |
| 10.0.0.24 | Uncertain  | 0.0000   | -     | Low      | Phase 1 | 1        | 11s      | 2026-09-25 13:09:48 | Attack Stopped                            |
| 10.0.0.17 | SYN Flood  | 0.6181   | 90.9% | Critical | Phase 2 | 1        | 11s      | 2026-09-25 13:09:46 | Attack persisted - escalating to time ban |
| 10.0.0.23 | ICMP Flood | 0.6434   | 94.5% | High     | Phase 2 | 1        | 11s      | 2026-09-25 13:09:45 | Attack persisted - escalating to time ban |
| 10.0.0.20 | UDP Flood  | 0.6448   | 77.9% | High     | Phase 2 | 1        | 10s      | 2026-09-25 13:09:44 | Attack persisted - escalating to time ban |
| 10.0.0.16 | SYN Flood  | 0.6474   | 91.8% | Critical | Phase 2 | 1        | 10s      | 2026-09-25 13:09:43 | Attack persisted - escalating to time ban |
| 10.0.0.16 | SYN Flood  | 0.6369   | 93.3% | Critical | Phase 2 | 2        | 1m 11s   | 2026-09-25 13:08:24 | Attack persisted - escalating to time ban |
| 10.0.0.24 | ICMP Flood | 0.6455   | 91.1% | High     | Phase 2 | 1        | 10s      | 2026-09-25 13:08:15 | Attack persisted - escalating to time ban |
| 10.0.0.10 | Uncertain  | 0.6129   | 68.7% | Low      | Phase 1 | 1        | 10s      | 2026-09-25 13:07:32 | Attack Stopped                            |
| 10.0.0.3  | Uncertain  | 0.5862   | 59.1% | Low      | Phase 1 | 1        | 11s      | 2026-09-25 13:07:31 | Attack Stopped                            |
| 10.0.0.4  | Uncertain  | 0.5950   | 59.1% | Low      | Phase 1 | 1        | 11s      | 2026-09-25 13:07:31 | Attack Stopped                            |
| 10.0.0.16 | SYN Flood  | 0.6456   | 90.4% | Critical | Phase 2 | 1        | 11s      | 2026-09-25 13:07:24 | Attack persisted - escalating to time ban |

| Source IP | Vector    | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                    |
|-----------|-----------|----------|-------|----------|---------|----------|----------|---------------------|-------------------------------------------|
| 10.0.0.18 | SYN Flood | 0.6162   | 93.1% | Critical | Phase 2 | 3        | 3m 11s   | 2026-09-25 13:07:01 | Attack persisted - escalating to time ban |
| 10.0.0.19 | SYN Flood | 0.6199   | 93.1% | Critical | Phase 2 | 3        | 3m 11s   | 2026-09-25 13:06:58 | Attack persisted - escalating to time ban |
| 10.0.0.17 | SYN Flood | 0.6087   | 93.3% | High     | Phase 2 | 3        | 3m 1s    | 2026-09-25 13:06:51 | Escalated to Time Ban                     |
| 10.0.0.21 | UDP Flood | 0.6448   | 83.4% | High     | Phase 2 | 3        | 3m 1s    | 2026-09-25 13:06:48 | Escalated to Time Ban                     |
| 10.0.0.22 | UDP Flood | 0.6448   | 82.5% | High     | Phase 2 | 3        | 3m 1s    | 2026-09-25 13:06:48 | Escalated to Time Ban                     |
| 10.0.0.18 | SYN Flood | 0.6162   | 93.1% | Critical | Phase 2 | 2        | 1m 11s   | 2026-09-25 13:05:01 | Attack persisted - escalating to time ban |
| 10.0.0.19 | SYN Flood | 0.6199   | 93.1% | Critical | Phase 2 | 2        | 1m 11s   | 2026-09-25 13:04:58 | Attack persisted - escalating to time ban |
| 10.0.0.17 | SYN Flood | 0.6087   | 93.3% | High     | Phase 2 | 2        | 1m 1s    | 2026-09-25 13:04:51 | Escalated to Time Ban                     |
| 10.0.0.21 | UDP Flood | 0.6448   | 83.4% | High     | Phase 2 | 2        | 1m 1s    | 2026-09-25 13:04:48 | Escalated to Time Ban                     |
| 10.0.0.22 | UDP Flood | 0.6448   | 82.5% | High     | Phase 2 | 2        | 1m 1s    | 2026-09-25 13:04:48 | Escalated to Time Ban                     |
| 10.0.0.18 | SYN Flood | 0.6274   | 93.1% | Critical | Phase 2 | 1        | 11s      | 2026-09-25 13:04:01 | Attack persisted - escalating to time ban |
| 10.0.0.2  | Uncertain | 0.6053   | 58.9% | Low      | Phase 1 | 1        | 10s      | 2026-09-25 13:03:58 | Attack Stopped                            |
| 10.0.0.19 | SYN Flood | 0.6278   | 93.1% | Critical | Phase 2 | 1        | 11s      | 2026-09-25 13:03:58 | Attack persisted - escalating to time ban |

## Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

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SIGNATURE OVER PRINTED NAME

**Network Administrator**

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DATE